```
'Ke'Shawn Stafford
'Project 10
'04/15/2025

' {$STAMP BS2}
' {$PBASIC 2.5}

C_Rev      PIN 11

C1         VAR Word
C2         VAR Byte
C3         VAR Nib
C4         VAR Byte
pass_Start VAR Nib(5)
password   VAR Nib(5)

#SELECT $STAMP
#CASE BS2, BS2E, BS2PE
baud CON 119 + 32768
#ENDSELECT

red        VAR Word
grn        VAR Word
blu        VAR Word
time       VAR Word
row        VAR Nib
column     VAR Nib
keypad     VAR Word
temp       VAR Nib

GOSUB Reset

SEROUT 11, baud, ["= (00 $ m) !"]
FOR C4 = 0 TO 23
PULSOUT 15, 255
PAUSE 20
NEXT
PAUSE 200
C1 = 0
SEROUT 14, 84, [22, 12, 17, 208]
SEROUT 14, 84, [128, "Enter Password"]
GOTO Start

Start:
    SEROUT 14, 84, [148]
    DO
    GOSUB C_Pal
    GOSUB K_Pad
    IF C2 = 6 THEN
    C2 = 0
    SEROUT 14, 84, [22, 12, 17, 208]
    SEROUT 14, 84, [128, "System is Armed"]
    SEROUT 14, 84, [218, 230]
    SEROUT 14, 84, [218, 229]
    PAUSE 1500
    GOTO Armed
    ENDIF
    LOOP
    RETURN

Armed:
    LOW 12
    HIGH 13
    C2 = 0
    C1 = 1
    SEROUT 14, 84, [12, 17, 208]
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    PAUSE 10
    SEROUT 14, 84, [128, "Door is Locked", 148]
    SEROUT 14, 84, [218, 227]
    SEROUT 14, 84, [218, 226]
    SEROUT 14, 84, [218, 225]
    DO
    GOSUB C_Pal
    GOSUB K_Pad
    IF C2 = 6 THEN
    GOSUB P_Check
    C2 = 0
    FOR C4 = 0 TO 25
    PULSOUT 15, 1120
    PAUSE 25
    NEXT
    PAUSE 200
    GOTO Disarm
    ENDIF
    LOOP
    RETURN

Disarm:
    HIGH 12
    LOW 13
    C1 = 2
    C4 = 0
    SEROUT 14, 84, [12, 17, 208]
    PAUSE 10
    SEROUT 14, 84, [128, "Door is Open", 148]
    SEROUT 14, 84, [218, 220]
    SEROUT 14, 84, [218, 223]
    SEROUT 14, 84, [218, 228]
    DO
    GOSUB C_Pal
    GOSUB K_Pad
    IF (C4 = 20) THEN
    FOR C4 = 0 TO 25
    PULSOUT 15, 255
    PAUSE 25
    NEXT
    PAUSE 200
    SEROUT 14, 84, [12, 17, 208]
    GOTO Armed
    PAUSE 10
    SEROUT 14, 84, [128, "Door is Open", 148]
    ELSEIF (C4 < 20) THEN
    C4 = C4 + 1
    ENDIF
    IF C2 = 6 THEN
    GOSUB P_Check
    C2 = 0
    C4 = 0
    PAUSE 200
    GOTO Armed
    ENDIF
    LOOP
    RETURN

C_Pal:
    SEROUT 11, baud, ["c"]
    PAUSE 100
    SERIN 11, baud, [WAIT("$"), HEX3 red, HEX3 grn, HEX3 blu]
    'DEBUG "R=", DEC red, " G=", DEC grn, " B=", DEC blu, CR
    GOSUB Colors
    PAUSE 100
    RETURN
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Check:
    IF (C1 = 0) THEN
    FOR C4 = 0 TO C2
    IF (pass_Start(C4) = C3) THEN
    SEROUT 14, 84, [148, " "]
    FOR C4 = 0 TO 5
    pass_Start(C4) = 0
    NEXT
    C2 = 0
    GOTO Start
    ENDIF
    NEXT
    ELSE
    FOR C4 = 0 TO C2
    IF (password(C4) = C3) THEN
    SEROUT 14, 84, [148, " "]
    FOR C4 = 0 TO 5
    password(C4) = 0
    NEXT
    C2 = 0
    IF C1 = 1 THEN
    GOTO Armed
    ELSE
    GOTO Disarm
    ENDIF
    ENDIF
    NEXT
    ENDIF
    RETURN

P_Check:
    C3 = 0
    C4 = 0
    FOR C4 = 0 TO 4
    IF (pass_Start(C4) = password(C4)) THEN
    C3 = C3 + 1
    PAUSE 20
    ELSE
    SEROUT 14, 84, [12]
    SEROUT 14, 84, [128, "Wrong Code Try Again"]
    PAUSE 2000
    C4 = 0
    C2 = 0
    IF C1 = 1 THEN
    GOTO Armed
    ELSE
    GOTO Disarm
    ENDIF
    ENDIF
    NEXT
    RETURN

Reset:
    LOW 11
    INPUT 11
    DO UNTIL 11 : LOOP
    LOW 11
    PAUSE 80
    INPUT 11
    PAUSE 10
    RETURN

K_Pad:
    GOSUB ReadKeyPad
    IF keypad THEN
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    GOSUB Display
    ENDIF
    RETURN

ReadKeyPad:
    keypad = 0
    OUTL = %00000000
    DIRL = %00000000
    FOR row = 0 TO 3
    DIRB = %1111
    OUTB = %0000
    OUTA = 1 << row
    DIRA = 1 << row
    temp = 0
    FOR column = 0 TO 3
    INPUT (column + 4)
    temp = temp | (INB & (1 << column))
    NEXT
    keypad = keypad << 4 | (Temp REV 4)
    NEXT
    RETURN

Display:
    IF keypad.BIT15 THEN
    SEROUT 14, 84, ["1", 218, 220]
    C3 = 1
    GOSUB Check
    GOSUB Plus
    ELSEIF keypad.BIT14 THEN
    SEROUT 14, 84, ["2", 218, 221]
    C3 = 2
    GOSUB Check
    GOSUB Plus
    ELSEIF keypad.BIT13 THEN
    SEROUT 14, 84, ["3", 218, 222]
    C3 = 3
    GOSUB Check
    GOSUB Plus
    ELSEIF keypad.BIT11 THEN
    SEROUT 14, 84, ["4", 218, 223]
    C3 = 4
    GOSUB Check
    GOSUB Plus
    ELSEIF keypad.BIT10 THEN
    SEROUT 14, 84, ["5", 218, 224]
    C3 = 5
    GOSUB Check
    GOSUB Plus
    ELSEIF keypad.BIT9 THEN
    SEROUT 14, 84, ["6", 218, 225]
    C3 = 6
    GOSUB Check
    GOSUB Plus
    ENDIF
    RETURN

Colors:
    IF (red > 20 AND red < 35 AND grn > 250 AND grn < 275 AND blu > 115 AND blu < 145) THEN
    SEROUT 14, 84, [148, "Green   ", 218, 220]
    C3 = 1
    GOSUB Check
    GOSUB Plus
    ELSEIF (red > 5 AND red < 20 AND grn > 180 AND grn < 220 AND blu > 365 AND blu < 385) THEN
    SEROUT 14, 84, [148, "Blue   ", 218, 221]
    C3 = 2
    GOSUB Check
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    GOSUB Plus
    ELSEIF (red > 290 AND red < 310 AND grn > 75 AND grn < 95 AND blu > 45 AND blu < 60) THEN
    SEROUT 14, 84, [148, "Red   ", 218, 222]
    C3 = 4
    GOSUB Check
    GOSUB Plus
    ELSEIF (red > 315 AND red < 350 AND grn > 510 AND grn < 545 AND blu > 120 AND blu < 145) THEN
    SEROUT 14, 84, [148, "Yellow   ", 218, 223]
    C3 = 3
    GOSUB Check
    GOSUB Plus
    ELSEIF (red > 130 AND red < 160 AND grn > 90 AND grn < 110 AND blu > 45 AND blu < 60) THEN
    SEROUT 14, 84, [148, "Brown   ", 218, 224]
    C3 = 5
    GOSUB Check
    GOSUB Plus
    ELSEIF (red > 270 AND red < 320 AND grn > 135 AND grn < 160 AND blu > 210 AND blu < 250) THEN
    SEROUT 14, 84, [148, "Pink   ", 218, 225]
    C3 = 6
    GOSUB Check
    GOSUB Plus
    ENDIF
    RETURN

Plus:
    IF (C1 = 0) THEN
    pass_Start(C2) = C3
    ELSE
    password(C2) = C3
    ENDIF
    C2 = C2 + 1
    PAUSE 500
    RETURN
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